Evaluating Leveraged DeFi Long–Short Positions under Jump Risk: Killed Moments, Liquidation Recovery and the Limits of Optimal Sizing

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DeFi long–short sizing under liquidation and jump risk

DeFi Lending Platforms

Aave / Compound: pooled lending

- **Mechanism** lenders supply pool assets; borrowers draw pooled liquidity
- **Accounting** no bilateral lender; protocol-level collateral and debt balances
- **Interest accrual** variable supply/borrow APRs; utilization-driven index growth
- **No fixed tenor** open-ended loans; no maturity date; no term structure contract

Overcollateralization

- Pseudonymous parties; no credit underwriting; no legal margin call
- Solvency and obligation enforcement entirely collateral-based
- No-arbitrage discipline through collateral constraint
- **Liquidator role** repay unsolvent debt; seize collateral bonus; restore pool solvency

Health Factor and Boundary

Key mechanism variables

State variables:

$$V_{1,t} := n_1 S_{1,t} \quad \text{collateral value}, \qquad V_{2,t} := n_2 S_{2,t} \quad \text{borrow/debt value}$$

Loan-to-value* and liquidation threshold:

$$\mathrm{LTV}_t := \frac{V_{2,t}}{V_{1,t}}, \qquad b = \text{threshold LTV}$$

Aave origination LTV $<$ threshold LTV b; initial buffer by design

Health factor:

$$H_t := \frac{b\, V_{1,t}}{V_{2,t}} = \frac{b}{\mathrm{LTV}_t}$$

Liquidation triggering condition:

$$H_t < 1 \quad \Longleftrightarrow \quad V_{2,t} > b\, V_{1,t}$$

* Remark: for now, ignore origination-LTV / threshold-LTV gap

Constructing the Long–Short Position

Step	Protocol action	Economic exposure
0	Initial capital injection: \$1-worth BTC	External capital only
1	Supply BTC collateral to Aave	Long BTC collateral
2	Borrow \$0.8-worth ETH	Short ETH liability
3	Sell ETH into \$0.8-worth BTC	Self-financing conversion
4	Re-supply BTC as collateral	Total BTC collateral: \$1.8
Result	Collateralized debt position	Leveraged long BTC, short ETH

Resulting normalized position:

- External capital: \$1; no additional capital after step 0
- BTC collateral: \$1.8; ETH debt: \$0.8
- $b = 0.8$; $H_0 = 0.8 \times 1.8/0.8 = 1.8$
- Leveraged BTC exposure; short ETH exposure

Looping the Position

Repeat steps 1–4; zero transaction cost and slippage

Round-m additional collateral:

$$1, \quad b, \quad b^2, \quad \ldots, \quad b^m$$

Total BTC collateral after m loops:

$$V_1^{(m)} = 1 + b + \cdots + b^m = \frac{1 - b^{m+1}}{1 - b}$$

ETH debt:

$$V_2^{(m)} = b + b^2 + \cdots + b^{m+1} = b\, V_1^{(m)}$$

Infinite-loop limit:

$$V_1^{(\infty)} = \frac{1}{1 - b}, \qquad V_2^{(\infty)} = \frac{b}{1 - b}$$

Example $b = 0.8$:

$$V_1^{(\infty)} = 5, \qquad V_2^{(\infty)} = 4, \qquad H_0^{(\infty)} = \frac{b V_1^{(\infty)}}{V_2^{(\infty)}} = 1$$

Extreme leverage: any adverse move $\rightarrow$ liquidation; favorable move $\rightarrow$ amplified gain

Unwinding and Taking Profit

Favorable scenario: BTC $+10\%$; ETH -16.7%

$$H_t = \frac{0.80\, V_{1,t}}{V_{2,t}}, \qquad \text{equity} = V_{1,t} - V_{2,t}$$

Scenario	BTC collat.	ETH debt	H	Equity	Action
Entry	\$1.80	\$0.80	1.80	\$1.00	Hold
Favorable move	\$1.98	\$0.667	2.38	\$1.313	Voluntary unwind

Outcome:

$$\text{gain} = 1.313 - 1.000 = 0.313$$

Voluntary unwind; solvent position; profit realization

Aave Liquidation and Immediate Risk

Adverse scenario: BTC -25%; ETH $+40\%$

$$H_t = \frac{bV_{1,t}}{V_{2,t}} = \frac{0.80\, V_{1,t}}{V_{2,t}}, \qquad H_t \le 1 \iff \frac{V_{2,t}}{V_{1,t}} \ge 0.80$$

Scenario	BTC collat.	ETH debt	H	Equity	Outcome
Entry	\$1.80	\$0.80	1.80	\$1.00	Safe
Adverse move	\$1.35	\$1.12	0.96	\$0.23	Liquidatable

Outcome:

$$H_t < 1 \quad \Rightarrow \quad \text{keeper liquidation; collateral seized at bonus}$$

Recovery drivers: shock severity; market sentiment/liquidity; liquidator inventory willingness; gas fees

Modeling convention: detailed recovery omitted; $\Pi_T = 0$ whenever $\tau \le T$

Payoff Function from the Trade

Holdings: n_1 units of collateral asset S_1; debt: n_2 units of borrowed asset S_2

Surviving-path terminal wealth:

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})\, \mathbf{1}_{\{\tau > T\}}$$

Running BTC/ETH example:

$$S_1 = \mathrm{BTC}, \qquad S_2 = \mathrm{ETH}, \qquad n_1 S_{1,0} = 1.80, \qquad n_2 S_{2,0} = 0.80$$

Therefore:

$$\Pi_T = \left(1.80\, \frac{S_{1,T}}{S_{1,0}} - 0.80\, \frac{S_{2,T}}{S_{2,0}}\right) \mathbf{1}_{\{\tau > T\}}$$

Survival indicator: $\tau \leq T \Rightarrow \Pi_T = 0$

Initial net book value normalized to one unit of account; moments stated directly in Π_T

Why Payoff Moments Are Hard

Target statistics:

$$\mathbb{E}[\Pi_T], \qquad \mathrm{Var}(\Pi_T), \qquad \mathbb{E}[\Pi_T^k], \quad k \geq 1$$

Killed payoff:

$$\Pi_T^k = (n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}$$

Terminal payoff state:

$$(S_{1,T}, S_{2,T})$$

Pathwise killing process:

$$\tau > T \quad \Longleftrightarrow \quad H_s \geq 1 \ \text{ for all } s \leq T, \qquad H_s = \frac{b\, n_1 S_{1,s}}{n_2 S_{2,s}}$$

Direct moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \mathbb{E}_{\mathbb{P}}\left[(n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}\right]$$

Difficulty: terminal bivariate payoff + pathwise barrier

Generic numerical routes:

- Monte Carlo; barrier monitoring / possible 2D Brownian-bridge refinement
- Numerical PDE/PIDE; two terminal state variables plus killing boundary

Literature Positioning

Intersection:

$$\text{DeFi liquidation} \quad + \quad \text{first-passage methods for jump diffusions}$$

- **DeFi liquidation** Qin et al. [4]; Heimbach–Huang [1]
- Incentive structure; leverage amplification; liquidation cascades
- Empirical focus; no ex-ante sizing rule

- **First passage under jumps** Kou–Wang [3]; Sepp [5]
- Double-exponential jumps; Laplace transforms; finite algebraic systems
- Classical object: single-asset barrier option

Gap:

$$\text{barrier-killed long–short ratio} \quad \neq \quad \text{single-asset barrier option}$$

Bridge: change of measure $\rightarrow$ one-dimensional killed expectation

Price Process

Jump motivation: crypto gaps; flash crashes; barrier crossed before rebalance Kou model [2]

Price dynamics, $i = 1, 2$:

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \qquad dX_{i,t} = \mu_i^X \, dt + \sigma_i \, dB_{i,t} + d\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)} \right)$$

Brownian correlation:

$$d\langle B_1, B_2 \rangle_t = \rho \, dt,$$

Poisson intensity: λ_i

Kou double-exponential jump density:

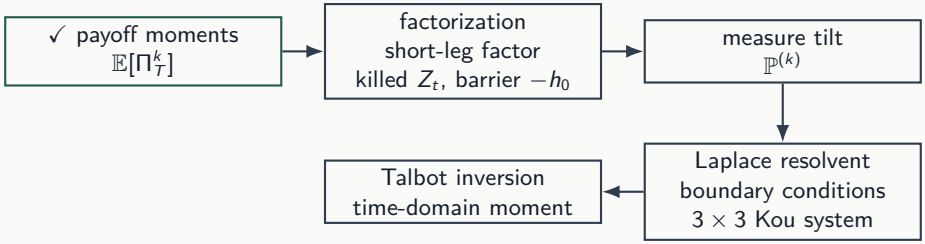
$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}$$

Drift normalization: $\mathbb{E}_{\P}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$

$$\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i \chi_i, \qquad \chi_i = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1$$

Constant supply/borrow rates absorbed into drift; dynamic rate modeling outside scope

Semi-Analytic Architecture



Covered: payoff function + need for moments

Next: factorization and measure change → one-dimensional killed expectation

$$k\,\eta_{2,+} < 1 \quad \text{moment-admissibility condition}$$

Factorization: From Two Assets to One Barrier

Motivation: separate terminal short-leg factor; isolate killed ratio problem

Bivariate payoff; scalar liquidation state:

$$Z_t = X_{1,t} - X_{2,t}$$

Initial log health factor:

$$h_0 := \log H_0$$

Factorization using $H_T = e^{h_0 + Z_T}$:

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} (e^{h_0 + Z_T} - b) \mathbf{1}_{\{\tau > T\}}$$

One-dimensional payoff kernel:

$$g_k(z; h_0) := (e^{h_0 + z} - b)^k,$$

Integer moments:

$$\Pi_T^k = \underbrace{\left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{k X_{2,T}}}_{\mathcal{F}_T^{S_2} \text{ part}} \underbrace{g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}}_{\mathcal{F}_T^Z \text{ killed-ratio part}}$$

Measure change: absorb $\mathcal{F}_T^{S_2}$ factor; retain killed expectation under Z

Barrier-option analogue: scalar state Z, barrier $-h_0$

Measure Change

Start from factorized moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \, \mathbb{E}_{\mathbb{P}}\left[e^{kX_{2,T}} g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Exponential tilt:

$$\left.\frac{d\mathbb{P}^{(k)}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = \frac{e^{kX_{2,T}}}{\mathbb{E}[e^{kX_{2,T}}]}$$

Heuristic reminder, with $Y = g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}$:

$$\mathbb{E}_{\mathbb{P}^{(k)}}[Y] = \int Y \, d\mathbb{P}^{(k)} = \int Y \, \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} \, d\mathbb{P}$$

$$= \mathbb{E}_{\mathbb{P}}\left[Y \, \frac{e^{kX_{2,T}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]}\right] \qquad = \frac{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y]}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]} \quad \Longrightarrow \quad \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y] = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]\mathbb{E}_{\mathbb{P}^{(k)}}[Y]$$

Remark: Radon–Nikodym theorem / change-of-measure identity

Reduced moment:

$$\mathbb{E}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \mathbb{E}[e^{kX_{2,T}}] \, \mathbb{E}^{(k)}\left[g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Bivariate payoff moment → killed expectation of the ratio process

Tilted Ratio Process

Remaining object:

$$\mathbb{E}^{(k)}\left[g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Question:

$$Z_t = X_{1,t} - X_{2,t} \quad \text{under} \quad \mathbb{P}^{(k)}$$

Under $\mathbb{P}^{(k)}$: Z remains a Lévy process

Lévy–Khintchine exponent of Z under $\mathbb{P}^{(k)}$:

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \frac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big)$$

with

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1 - p_i}{1 + s\eta_{i,-}}, \qquad \sigma_Z^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

$\mathcal{L}^{(k)}$: infinitesimal generator of Z under $\mathbb{P}^{(k)}$, the generator eigenrelation:

$$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$$

Killed Moment PIDE

Target function:

$$U(t,z) := \mathbb{E}_z^{(k)}\left[g_k(Z_t;h_0)\mathbf{1}_{\{\tau(h_0)>t\}}\right], \qquad z > -h_0$$

Backward Kolmogorov PIDE for the killed process:

$$\partial_t U(t,z) = \mathcal{L}^{(k)} U(t,z), \qquad z > -h_0$$

Initial and absorbing boundary:

$$U(0,z) = g_k(z;h_0), \qquad U(t,z) = 0 \quad \text{for } z \le -h_0$$

One time variable; one state variable z for the ratio process Z

Laplace Transform Trick

Time-domain PIDE:

$$\partial_t U(t,z) = \mathcal{L}^{(k)} U(t,z), \qquad U(0,z) = g_k(z;h_0)$$

Laplace transform in time:

$$\widehat{U}(q,z) = \int_0^\infty e^{-qt} U(t,z)\, dt, \qquad q > 0$$

Laplace-domain resolvent:

$$(q - \mathcal{L}^{(k)})\widehat{U}(q,z) = g_k(z;h_0), \qquad z > -h_0$$

Sepp [5] idea:

Laplace transform + double-exponential jumps $\rightarrow$ finite algebraic system

Here:

$$\psi_Z^{(k)}(\gamma) = q \quad \Longrightarrow \quad \text{degree-six characteristic equation}$$

Three admissible roots + three boundary conditions $\rightarrow 3 \times 3$ system

Resolvent Solution Form

Resolvent decomposition:

$$\widehat{U}(q, z; h_0) = \widehat{U}_{\mathrm{part}}(q, z; h_0) + \widehat{U}_{\mathrm{hom}}(q, z; h_0)$$

Particular solution:

$$\widehat{U}_{\mathrm{part}}(q, z; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} e^{jz}$$

Homogeneous solution:

$$\widehat{U}_{\mathrm{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)(z+h_0)}$$

Boundary coefficients:

$$\underbrace{\mathbf{M}_{\mathrm{bar}}^{(k)}(q)}_{3\times 3} \underbrace{\mathbf{B}^{(k)}}_{3\times 1} = -\underbrace{\mathbf{b}^{(k)}}_{3\times 1}$$

PIDE solve $\rightarrow$ summation of exponentials + one 3×3 linear system

Benchmark: Drift Kept Outside the Fit

ECF on 840 four-hour WBTC/WETH returns (Aave v3, 2026-03-17 to 08-04): $\sigma_i, \lambda_i, p_i, \eta_{i,\pm}, \rho$

Drift is not fitted

- Drift precision depends only on calendar length, not on sampling frequency
- A free drift in the ECF phase on de-meaned returns = normalizer, not forecast ($\mu_1 - \mu_2 = -0.86$, wrong sign)
- Benchmark: window's sample drift incl. Aave rates, $\mu_1 = -0.314$, $\mu_2 = -0.417$
- Spread drift $\nu = \mu_1 - \mu_2 = +0.103$/yr ($+0.128$ price, -0.025 net carry)

Any stated view can be substituted — the map is exact given its inputs

Validation Against Monte Carlo

h_0	H_0	$p_{\mathrm{surv}}^{\mathrm{Lap}}$	$p_{\mathrm{surv}}^{\mathrm{MC}}$	Mean	Var.
0.07	1.07	0.6053	0.6190	1.1386	0.0387
0.10	1.11	0.7545	0.7628	1.0840	0.0367
0.25	1.28	0.9906	0.9912	0.9914	0.0282
0.50	1.65	1.0000	1.0000	0.9817	0.0149
1.00	2.72	1.0000	1.0000	0.9776	0.0085
2.00	7.39	1.0000	1.0000	0.9752	0.0068
∞	∞	1	—	0.9742	0.0065

Conditional moments; no-shorting limit exact to 10^{-12}

Survival at $h_0 = 0.07$ (Laplace 0.6053)

steps	bridge MC	naive bias
125	0.6077	+0.0145
500	0.6061	+0.0076
4,000	0.6065	+0.0039

- Naive MC misses intra-step crossings: one-sided bias
- Bridge-corrected: within 1 s.e. from 500 steps
- Laplace 23 ms vs $\approx$ 22 s: three orders of magnitude, no noise

Liquidation Recovery and Jump Overshoot

Liquidator repays debt, seizes collateral worth $(1+\ell)\times$ repayment

$$R_\tau = V_{2,\tau}\left(\frac{e^{h_\tau}}{b} - (1+\ell)\right), \qquad h_\tau \le 0$$

- Exact via *bivariate tilts* of the same 3×3 system: $R_\tau = Ae^{X_{1,\tau}} - Be^{X_{2,\tau}}$
- Recovery matters: at $H_0 = 1.16$, $\mathbb{E}[W_T]$ $0.927 \to 0.981$, variance $0.127 \to 0.053$
- Overshoot correction -3% at fitted (small) jumps
- Scaled jumps: -59% at $10\times$; at $20\times$ mean recovery < 0 (bad debt)
- Same at *fixed* jump variance: jump size, not variance, drives it

Overshoot Law, Bad Debt and Log Wealth

Overshoot $u = -h_\tau$: creeping atom + two exponential phases

$$\mathbb{E}[f(u)\mathbf{1}_{\{\tau \le T\}}] = P_0 f(0) + \sum_{i=1}^{2} P_i \int_0^\infty f(u)\, r_i e^{-r_i u} \mathrm{d}u, \quad r_1 = \tfrac{1}{\eta_{1,-}},\ r_2 = \tfrac{1}{\eta_{2,+}}$$

Weights from three hitting moments; checked at $j = 5, 10$ to 10^{-10}

Bad debt in closed form: $\mathbb{P}(R_\tau \le 0, \tau \le T) = P_1 e^{-r_1 u^\star} + P_2 e^{-r_2 u^\star}$

- 10^{-9} at the headline fit; 3×10^{-4} under the full-history fit (larger jumps)

Expected log wealth (Kelly) semi-analytic

$$\log(e^h - b) = h - \sum_{n \ge 1} \tfrac{b^n}{n} e^{-nh} \ \Rightarrow\ \mathbb{E}[\log W_T] = \text{tilted killed moments} + \text{overshoot integral}$$

- Agrees with bridge MC within 1.6 s.e.; ≈ 0.3 s per buffer

Decision Rules on the Map

Rule	Parameter	L_0
Liquidation budget	$p_{\text{liq}} \leq 0.05$	2.86
	$p_{\text{liq}} \leq 0.10$	3.05
	$p_{\text{liq}} \leq 0.20$	3.31
Mean–variance on W_T	$\alpha = 0.5$	2.19
	$\alpha = 1$	1.91
	$\alpha = 2$	1.40
Kelly	full	1.91
	half	1.45
	quarter	1.23
Any of these, no drift view	—	1

Benchmark sample drift $\nu = +0.103$; avoid conditional moments (survivorship)

- Budget rule robust: $L_0 \in [2.66, 3.04]$ for $\nu \in [-0.5, +0.6]$
- Mean–variance and Kelly driven by the drift
- No relative-drift view $\Rightarrow$ unlevered
- Fractional Kelly scales debt, not the decision to borrow
- Search must include unlevered: capping at $H_0 \leq 2$ manufactures an interior optimum

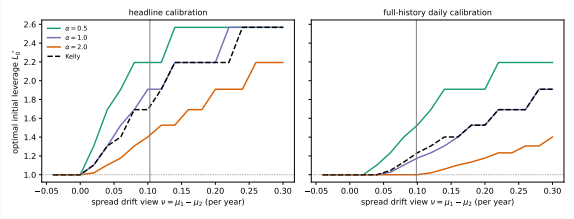
When Does Leverage Pay? The Break-Even Drift

First unit of debt $x \to 0$: $W_T = (1+x)e^{X_{1,T}} - xe^{X_{2,T}}$

$$\left.\frac{\mathrm{d}\,\mathbb{E}\log W_T}{\mathrm{d}x}\right|_0 = 1 - \mathbb{E}e^{X_{2,T}-X_{1,T}} \;\Rightarrow\; \nu^\star_{\mathrm{Kelly}} = \underbrace{\sigma_1^2 - \rho\sigma_1\sigma_2}_{\text{diffusion}<0} + \underbrace{\lambda_1\big[M_1(1)+M_1(-1)-2\big]}_{\text{WBTC jumps}>0}$$

Calibration	$\alpha = 1$	Kelly
Headline	0.008	0.008
Full, daily	0.054	0.048
2024–, 4h	0.049	0.045

- Independent of T; $\approx$ linear in α
- Small: 0.4–11%/yr
- Positive only via collateral's own jumps



Can the Data Tell?

$$\mathrm{sd}(\widehat{\mu}) = \frac{\sigma}{\sqrt{T}} \qquad \text{(calendar length only)}$$

- Sample spread drift $\hat{\nu} = +0.10$; s.e. 0.34 (headline), 0.19 (3.7 years)
- Separating $+0.10$ from $+0.05$ at 2 s.e. needs ≈ 210 years
- Posterior $\mathbb{P}(\nu > \nu^\star) \approx$ one half (carry-anchored prior); ≈ 0.6 (flat)
- Bayes–Kelly: lever iff $m > \nu^\star + \frac{1}{2} T s^2 \Rightarrow$ decided by the prior's shrinkage
- Reversed book (WETH collateral): $\nu = -0.11$ vs break-even $+0.10$ to $+0.19$

Optimal leverage hinges on an input the data cannot pin down

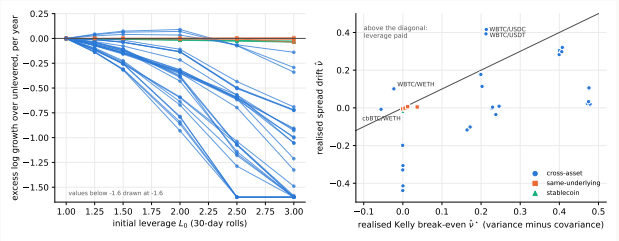
Did Leverage Pay? Realised Replay of 37 Pairs

Hourly replay, Aave v3 2023-01 to 2026-10, protocol liquidation rules, no model

Roll	$L_0 = 1.25$	$L_0 = 2$
1 day	−0.04 (4)	−0.29 (3)
30 days	−0.04 (4)	−0.33 (4)
Hold	−0.02 (12)	−0.30 (10)

Median excess log growth/yr (pairs where leverage paid)

- 1.25× beat unlevered in 4 of 37, none significant
- 2025: 0/29 cross-asset pair-years positive
- One cycle; BTC ×3.6, BTC/ETH ×2.3



What the Replay Says About the Model

WBTC/WETH, annualised

	Model (independent jumps)	Realised, daily
Correlation	0.50	0.81
Covariance	0.11	0.24
Spread variance	0.23	0.14
Break-even $\nu^\star = \mathrm{Var}_1 - \mathrm{Cov}$	$+0.048$	-0.023

- 85% of large days move both legs the same way: common jumps
- Independent jumps cannot carry co-movement $\Rightarrow$ spread variance overstated by 60%
- Explains the out-of-sample over-prediction of p_{liq} (1.3–1.6$\times$)
- Fix: a common Poisson component (stays in the exponential-phase class)

Interest-Rate Carry and Regimes

- Deterministic carry = per-leg drift shift $\mu_1 \mapsto \mu_1 + r_s$, $\mu_2 \mapsto \mu_2 + r_b$
- 37-pair pre-registered cross-section: carry worth ≤ 0.035 barrier s.d. per month for cross-asset pairs
- Rate-curve kink $\Rightarrow$ two-state regime, 6×6 resolvent
- Calibrated: opening regime moves survival by ≤ 0.09 pp at a month — inert
- Depeg d = same book at $h_0 + \log(1 - d)$: low-volatility pairs are depeg-fragile

Conclusions

- Leveraged DeFi long–short $\rightarrow$ one scalar first-passage problem in h_0
- Killed moments, survival, exact recovery, overshoot law, bad debt, $\mathbb{E}\log W$: one 3×3 system
- Validated; three orders of magnitude faster than simulation
- Sizing: any objective hinges on a small break-even drift the data cannot estimate
- Ex post, leverage rarely paid ($4/37$ pairs at $1.25\times$, none significant)
- Output is the map, not a leverage number

Next

- Common jump component
- Decision-theoretic sizing under drift uncertainty
- Endogenous utilization; dynamic collateral management

Thank you

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